

BASICS FOR AIR TRAFFIC CONTROL – FUNDAMENTALS OF WEATHER AND AVIATION WEATHER SERVICES

INTRODUCTION

Weather plays an integral role for controllers. Controllers are required to be familiar with, and stay familiar with, current weather information needed to perform Air Traffic Control (ATC) duties.

What do you, as an air traffic controller, need to know about weather?

Despite the development of advanced technologies such as radar and Global Positioning Systems (GPSs), safety in flight is still subject to hazardous weather. Thus, you must have a fundamental knowledge of weather to help pilots avoid hazardous weather.



This module will help you understand basic principles of weather, different elements of the atmosphere, and the National Weather Service's (NWS) role in providing aviation weather services to the FAA.

EARTH'S ATMOSPHERE

Purpose: This lesson will help you recognize characteristics associated with each layer of the Earth's atmosphere, and how they affect weather and aviation.

Objective:

- Identify characteristics and properties of the Earth's atmosphere

The references for this lesson are as follows:

- NWS: Jetstream: Online School for Weather
- FMH-1, Surface Weather Observations and Reports
- FAA-H-8083-28, Aviation Weather Handbook

Earth's Atmosphere

The atmosphere is a cloud of gas and suspended solids extending from the Earth's surface out thousands of miles. This gas grows thinner as the distance from the Earth increases.

The atmosphere:

- Surrounds the Earth
- Is held in place by gravitational pull
- Holds the air we breathe
- Protects us from outer space
- Holds moisture, clouds, gases, and tiny particles

This protective bubble **absorbs ultraviolet solar radiation**, which warms the surface through heat retention and reduces temperature extremes.



Atmosphere

The atmosphere is subdivided into the five layers shown here. This lesson will emphasize the troposphere and lower stratosphere. This is where virtually all aircraft fly.



Troposphere

The troposphere is the **lowest layer** of the Earth's atmosphere and **extends around 36,000 feet** out from the Earth's surface.

The troposphere is where:

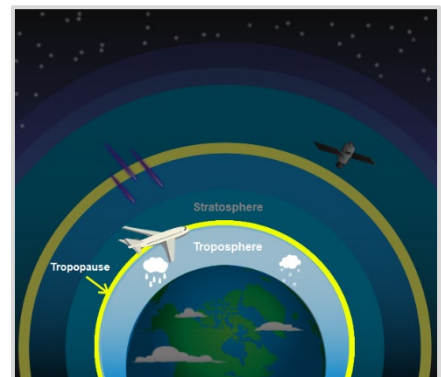
- Almost all **clouds and precipitation** occur
- **Most weather takes place**
- **Air pressure and density decrease with altitude**
- **Temperature generally decreases with altitude**

Note: An increase in temperature with altitude is abnormal and is defined as an **inversion**.

The **vertical depth varies** due to **fluctuating temperature associated with latitude and season**:

- At the equator, the depth is about 65,000'
- At 50°N and 50°S latitude, the depth is about 30,000'
- At the poles, the depth is about 20,000'

The **transition boundary between the troposphere and the stratosphere** is called the **tropopause**.

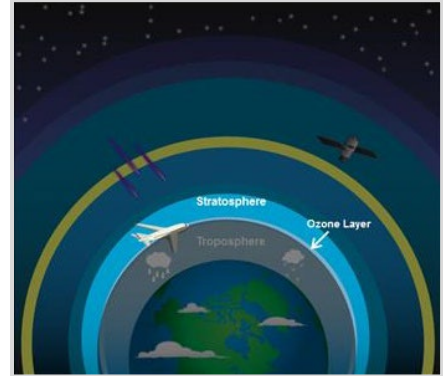


Stratosphere

The stratosphere extends from the tropopause up to 31 miles above the Earth's surface.

The stratosphere is where:

- Temperature increases with altitude, making it devoid of significant weather
- Thunderstorm tops reside, which commonly extend into the lower stratosphere
- The ozone layer resides



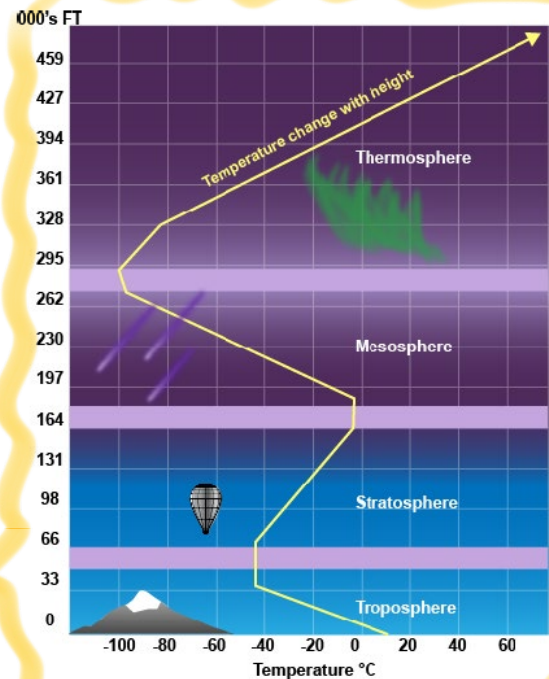
Standard Atmosphere

Standard Atmosphere is a hypothetical vertical distribution of the atmospheric temperature, pressure, and density, which, by international agreement, is considered to be representative of the atmosphere.

Standard atmosphere is used for:

- Pressure altimeter calibrations
- Aircraft performance calculations
- Aircraft design
- Weather-related processes

Property	Units
Sea level pressure	29.92 inches of mercury
Sea level temperature	15°C / 59°F°
Decrease of temperature with height (lapse rate) in the troposphere	2°C / 3.5°F / 1,000 feet
Pressure altitude of the tropopause	36,000 feet (FL 360)

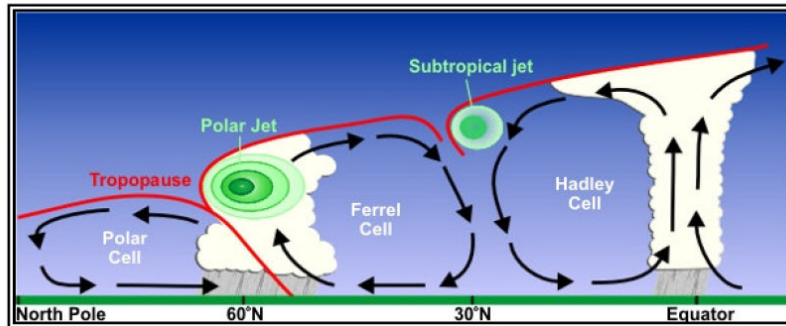


Jet Streams

Jet streams are **migrating streams of high-speed winds present at high altitudes**. Two jet streams that are commonly identified are the **polar jet stream** and the **subtropical jet stream**.

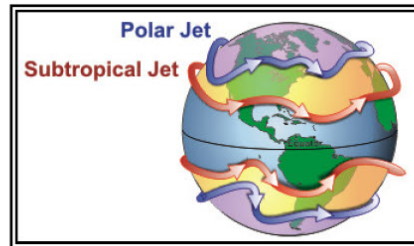
- 8.3.3 Location.** In addition, with the three cell circulations mentioned previously, the regions around 30° N/S and 50°-60° N/S are areas where temperature changes are the greatest. As the difference in temperature between the two locations increases, the strength of the wind increases. Therefore, the regions around 30° N/S and 50°-60° N/S are also regions where the wind in the upper atmosphere is the strongest.

Figure 8-4. Three Cell Circulations and Jet Stream Location



The 50°-60° N/S region is where the polar jet is located with the subtropical jet located around 30° N. Jet streams vary in height of 4 to 8 miles and can reach speeds of more than 275 miles per hour (239 knots /442 kilometers per hour).

Figure 8-5. Polar and Subtropical Jet Streams



The actual appearance of jet streams results from the complex interaction between many variables, such as the location of high and low pressure systems, warm and cold air, and seasonal changes. They meander around the globe, dipping and rising in altitude/latitude, splitting at times and forming eddies, and even disappearing altogether to appear somewhere else.

FEATURE	FEATURE/CHARACTERISTIC
Flow	The jet stream flows from west to east , but the flow often shifts to the north and south due to weather systems and seasonal changes.
Nature	Jet streams naturally extend around the globe but are segmented, splitting at times and even disappearing altogether to appear somewhere else.
Boundaries	Jet streams are associated with the boundaries between hot and cold air . Since these boundaries are most pronounced during winter, that is when jet streams are strongest.
ATC Challenges	Jet streams make air traffic control (ATC) separation more difficult. They often produce Clear Air Turbulence (CAT) and cause aircraft to move at different ground speeds.



Knowledge Check A

REVIEW what you have learned so far about the Earth's atmosphere. ANSWER the questions listed below.

1. Which layer of the atmosphere contains almost all clouds and precipitation? (Select the correct answer.)
 - ☐ Stratosphere
 - ☒ Troposphere
 - ☐ Mesosphere
2. Which layer of the atmosphere is stable and generally devoid of significant weather? (Select the correct answer.)
 - ☒ Stratosphere
 - ☐ Troposphere
 - ☐ Mesosphere
3. What is the value of sea level pressure in the standard atmosphere? (Select the correct answer.)
 - ☒ 29.92 inches of mercury
 - ☐ 29.92 hectopascals
 - ☐ 1013.2 inches of mercury
4. What is produced when a migrating stream of high-speed winds are present at high altitudes? (Select the correct answer.)
 - ☐ Vortices
 - ☒ Jet streams
 - ☐ Downwash
5. Which two jet streams are commonly identified? (Select the correct answer.)
 - ☐ Arctic and Tropical
 - ☐ Hot and Cold
 - ☒ Polar and Subtropical

Earth's Atmosphere Summary

The atmosphere is a bubble that surrounds Earth and protects the life within it. This lesson identified characteristics and properties of the Earth's atmosphere. Knowing how atmosphere can affect flight will allow you to provide better air traffic control services.

ATMOSPHERIC MOISTURE

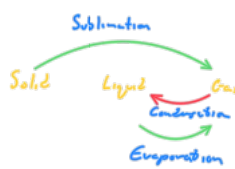
Purpose: This lesson explains the different characteristics of atmospheric moisture and the various forms of clouds.

Objectives:

- Describe characteristics of atmospheric moisture
- Define forms of clouds

The references for this lesson are as follows:

- FAA-H-8083-28, Aviation Weather Handbook
- NWS: Jetstream: Online School for Weather



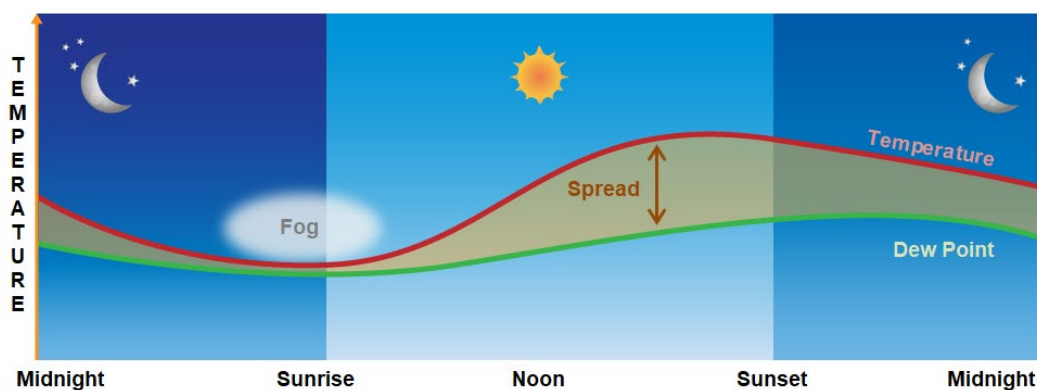
Atmospheric Moisture

Atmospheric moisture defines the amount of water vapor in the atmosphere.

TYPES	DEFINITION
Water Vapor	The gaseous form of water.
Evaporation	The phase transition by which a liquid is changed to a vapor (gas).
Sublimation	The phase transition by which a solid is changed into vapor (a gas) without passing through the liquid phase.
Temperature	A numerical value representing the average kinetic energy of the atoms and molecules within matter. Temperature depends directly on the energy of molecular motion. The air's capacity to hold water is directly related to its temperature. Warm air can hold more water vapor than cold air.
 Dew Point	The temperature an air parcel must be cooled at constant pressure and constant water vapor pressure to allow the water vapor in the parcel to condense into water (dew). <i>↑ Dew Point</i> Higher dew points at a location generally indicate higher quantities of water vapor. <i>↑ Water Vapor</i>
Condensation	The phase transition by which vapor (a gas) is changed into a liquid.
Relative Humidity	The ratio, usually expressed as a percentage, of water vapor actually in the air parcel compared to the amount of water vapor the air could hold at a particular temperature and pressure .
Saturation	The maximum possible quantity of water vapor that a parcel of air can hold at any given temperature and pressure. "Saturated" means an air parcel contains all the water vapor it can hold. "Unsaturated" means it has the capacity to hold more. 

Temperature/Dew Point Spread

The **difference between air parcel's temperature and its dew point temperature** is the **dew point depression**, or **commonly referred to as the spread**. During a typical 24-hour period over land, the surface air temperature increases during the day and decreases at night, while the dew point changes more slowly.





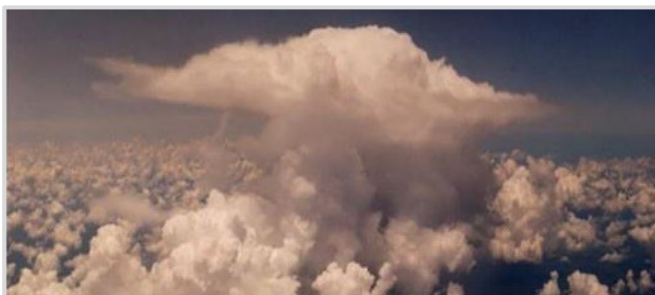
Knowledge Check B

REVIEW what you have learned so far about the atmospheric moisture. ANSWER the questions listed below.

- What is the name of the process in which a solid is changed into vapor without passing through the liquid phase? (Select the correct answer.)
 - ☒ Evaporation
 - ☐ Sublimation
 - ☐ Condensation
- What is the ratio of water vapor actually in the air parcel, compared to the amount of water vapor the air could hold at a particular temperature and pressure? (Select the correct answer.)
 - ☒ Relative humidity
 - ☐ Condensation
 - ☐ Sublimation
- The maximum possible quantity of water vapor that a parcel of air can hold at any given temperature and pressure is called _____. (Select the correct answer.)
 - ☐ Sublimation
 - ☐ Dew point
 - ☒ Saturation

Clouds

A cloud is a visible aggregate of minute water droplets and/or ice particles in the atmosphere above the Earth's surface. Clouds form when air is cooled to its dew point and becomes saturated. The most common way is via condensation of water vapor in rising air currents.

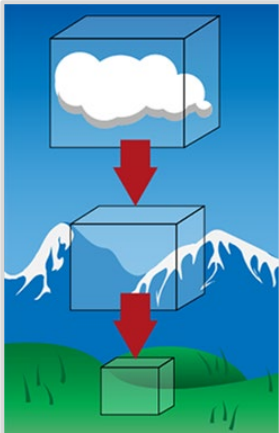


Cloud Formation

	6,000 feet Temperature 8°C Dew Point 8°C Spread 0°C (Saturated)	A parcel of rising air expands and cools as pressure decreases with altitude. As temperature/dew point spread decreases , a cloud will form, given sufficient lift to achieve saturation.
	3,000 feet Temperature 15°C Dew Point 12.5°C Spread 2.5°C (Unsaturated)	
	Surface Temperature 24°C Dew Point 14°C Spread 10°C (Unsaturated)	

Cloud
Forms ↑
↓ Cloud
Dissipates
(Rain)

Cloud Dissipation

	6,000 feet Temperature 8°C Dew Point 8°C Spread 0°C (Saturated)	A parcel of sinking air warms as it encounters increasing pressure and is compressed. As temperature/dew point spread increases , a cloud will quickly dissipate due to sinking air.
	3,000 feet Temperature 17°C Dew Point 9.5°C Spread 7.5°C (Unsaturated)	
	Surface Temperature 26°C Dew Point 11°C Spread 15°C (Unsaturated)	

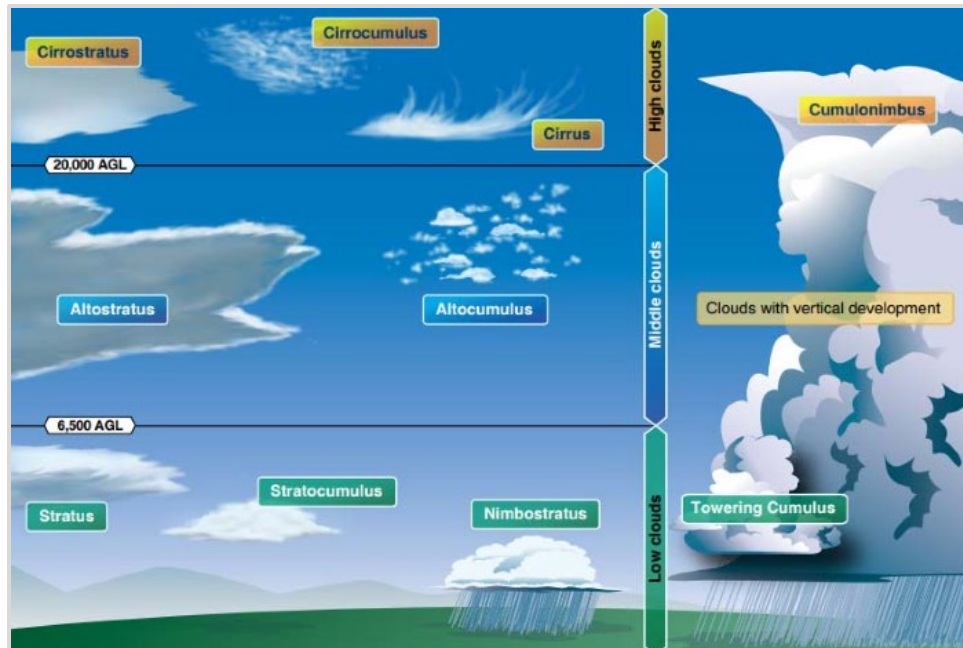
Cloud Forms

There are three forms of clouds: **cirriform**, **cumuliform**, and **stratiform**. The details are shown below.

 Cirriform	 <i>Nothing</i>	• High-level clouds which form above 20,000 feet • Usually composed of ice crystals • Typically thin and white in appearance • Contains no significant icing
 Cumuliform	 <i>Turb + Ice</i>	• Resembles white fluffy cotton balls or heaps • Indicates upward vertical motion or thermal uplift of air • Tops can reach over 60,000 feet • May produce icing, turbulence, and other hazards
 Stratiform	 <i>No Turb. Yes Ice</i>	• "Stratus" is Latin for "layer" or "blanket" • Consists of a featureless low layer that can cover sky • Often produces widespread Instrument Flight Rules (IFR) weather • Little or no turbulence but can produce icing

Cloud Prefixes and Suffixes

The prefix **nimbo-** or the suffix **-nimbus** means “raincloud.” Stratiform clouds from which rain is falling are **nimbostratus**, and cumuliform clouds which produce precipitation are **cumulonimbus**.



Knowledge Check C

REVIEW what you have learned so far about the clouds. ANSWER the questions listed below.

1. A parcel of _____ air expands and cools as pressure decreases with height. (Select the correct answer.)
 - ☒ Rising
 - ☐ Sinking
 - ☐ Stationary
2. Which cloud form indicates upward vertical motion or thermal uplift of air? (Select the correct answer.)
 - ☐ Cirriform
 - ☐ Stratiform
 - ☒ Cumuliform
3. Which cloud form often produces widespread IFR weather? (Select the correct answer.)
 - ☐ Cirriform
 - ☒ Stratiform
 - ☐ Cumuliform
4. Which type of cloud formations are rainclouds? (Select all correct answers that apply.)
 - ☐ Cirrinimbus
 - ☒ Cumulonimbus
 - ☒ Nimbostratus
 - ☐ Stratonimbus

Atmospheric Moisture Summary

The atmosphere, by nature, contains moisture which will affect flight in many different ways. This lesson explained the different characteristics of atmospheric moisture and the various forms of clouds. Knowing the details of atmospheric moisture and clouds will give you a better handle on what the pilots are dealing with and how you can help guide them.

WIND, AIR MASSES, AND FRONTS

Purpose: This lesson explains the details of air masses, mass modifications, and frontal types.

Objectives:

- Identify air masses and their characteristics
- Identify air mass modifications and their characteristics
- Identify frontal types and their characteristics

The references for this lesson are as follows:

- FAA-H-8083-28, Aviation Weather Handbook
- NWS: Jetstream: Online School for Weather

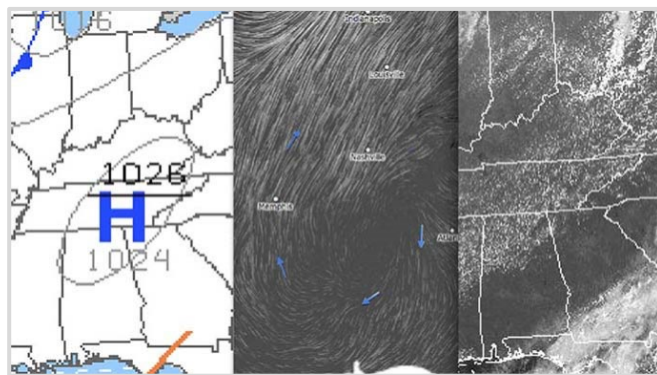
Wind

Wind is the air in motion relative to the surface of the Earth. The vertical component of the wind is typically very small compared to the horizontal component. However, it is very important in the formation and dissipation of clouds and precipitation.

High Weather Effects

A high occurs when a maximum of atmospheric pressure in two dimensions (closed isobars) on a surface chart, or a maximum of height (closed contours) on a constant pressure chart; also known as an anti-cyclone. Air flow around a high diverges in a clockwise motion and sinks.

- Sinking air compresses and warms
- Warming air can hold more water vapor so clouds tend to evaporate

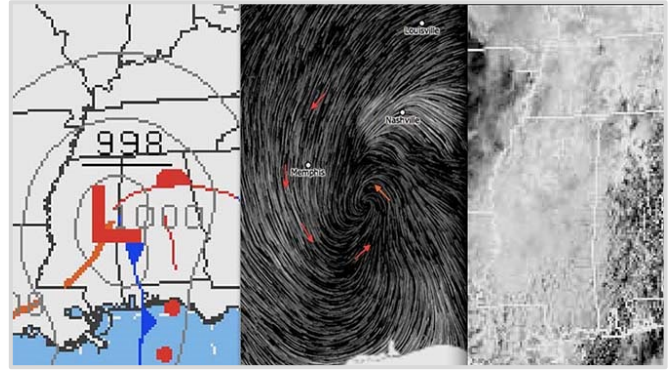


*High Pressure = Good Weather
Low Pressure = Bad Weather*

Low Weather Effects

Low occurs when there is a minimum of atmospheric pressure in two dimensions (closed isobars) on a surface chart, or a minimum of height (closed contours) on a constant pressure chart; also known as a cyclone. Air flow around a low converges in a counterclockwise motion and rises.

- Rising air expands and cools
- Cooling air can hold less water vapor so some of the invisible vapor may eventually condense into clouds and precipitation



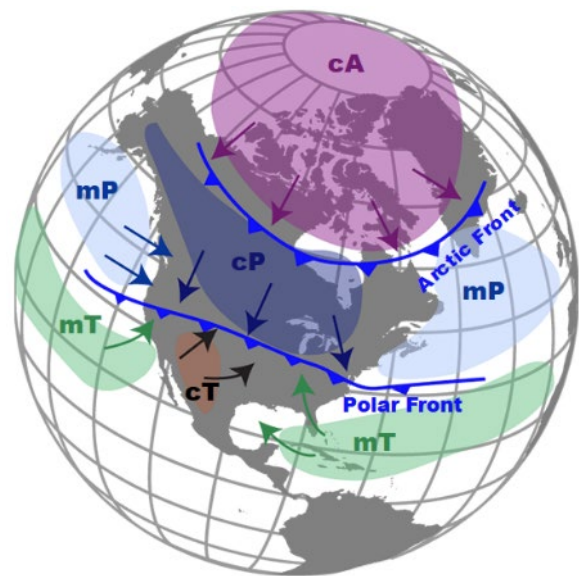
Knowledge Check D

REVIEW what you have learned so far about wind. ANSWER the questions listed below.

1. Which type of weather effect occurs when there is a minimum amount of atmospheric pressure on Earth's surface? (Select the correct answer.)
 - ☐ High
 - ☒ Low
 - ☐ Wind
2. Which type of weather effect occurs when there is a maximum amount of atmospheric pressure on Earth's surface? (Select the correct answer.)
 - ☒ High
 - ☐ Low
 - ☐ Wind

Air Masses

An **air mass** is a large body of air with generally uniform temperature and humidity. An **air mass source region** is a region where air masses originate and acquire their properties of temperature and moisture. These properties are acquired by prolonged contact (days to weeks) with the underlying surface. Air masses are classified according to the temperature and moisture properties of their source regions.



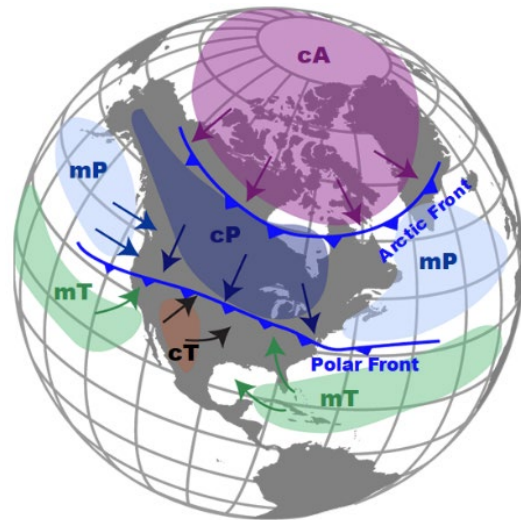
Air Mass Classification

Air masses are classified according to the temperature and moisture properties of their source regions.

TEMPERATURE	MOISTURE
Temperature properties ■ Arctic (A) – An extremely deep, cold air mass which develops mostly in winter over arctic surfaces of ice and snow ■ Polar (P) – A relatively shallow, cool-to-cold air mass which develops over high latitudes ■ Tropical (T) – A warm-to-hot air mass which develops over low latitudes	Moisture properties ■ Continental (c) – A dry air mass which develops over land ■ Maritime (m) – A moist air mass which develops over water

When this classification scheme is applied, the five air masses shown in the chart below may be identified.

Source Region	Continental (c)	Maritime (m)
Arctic (A)	Continental Arctic (cA) (Cold, dry)	Not Applicable
Polar (P)	Continental Polar (cP) (Cold, dry)	Maritime Polar (mP) (Cool, moist)
Tropical (T)	Continental Tropical (cT) (Hot, dry)	Maritime Tropical (mT) (Warm, moist)

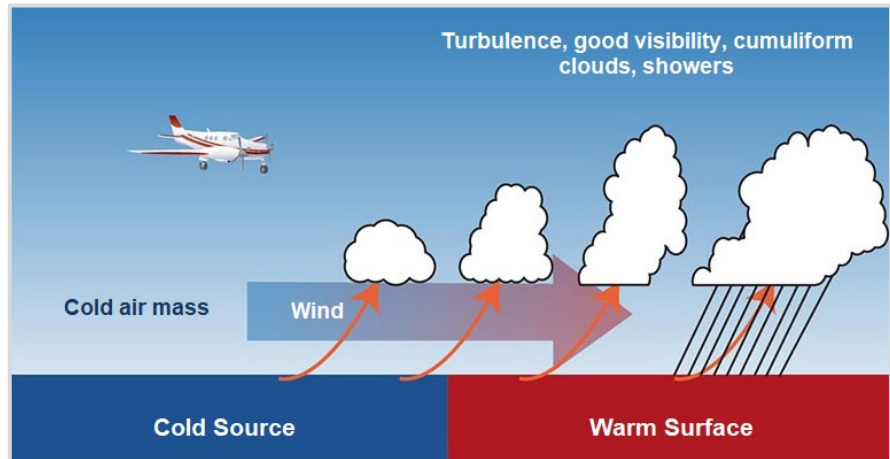


Note: Maritime Arctic (mA) is not listed since it seldom, if ever, forms due to the Arctic Ocean being frozen most of the time.

Air Mass Modification

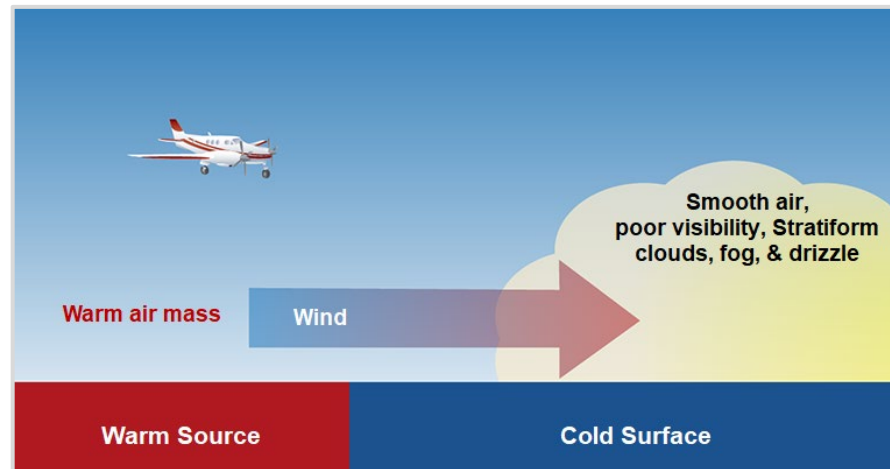
As an air mass is blown away from its source region, it is modified by its underlying surface. Cold air mass moving over a warm surface often produces unstable air associated with:

- Turbulence
- Good visibility (outside of clouds)
- Cumuliform clouds and showers



A warm air mass moving over a cold surface often produces stable air associated with:

- Smooth air
- Poor visibility
- Stratiform clouds, fog, and drizzle



Knowledge Check E

REVIEW what you have learned so far about air masses. ANSWER the questions listed below.

1. A Maritime Tropical (mT) air mass is _____ and _____. (Select the correct answer.)
 - ☒ Warm, moist
 - ☐ Warm, dry
 - ☐ Cold, dry
2. Cold air mass moving over a warm surface often produces unstable air associated with _____. (Select the correct answer.)
 - ☐ Smooth air
 - ☐ Poor visibility
 - ☒ Turbulence

Fronts

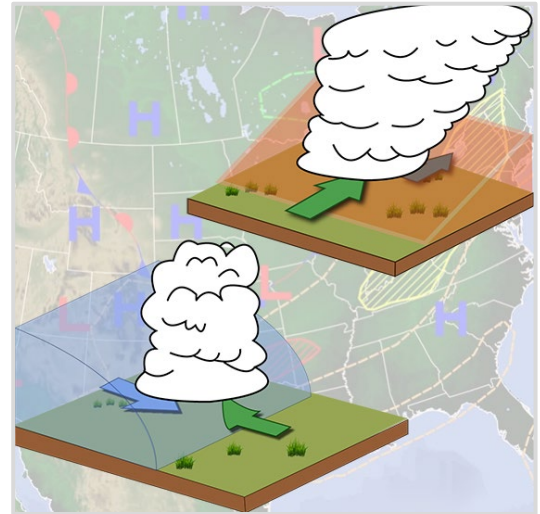
A front is a boundary or transition zone between two air masses of different density and usually different temperatures. Fronts are normally detectable at the surface when:

- There are significant temperature differences along frontal zones
- Winds come together at the fronts
- Pressure decreases as a front approaches and increases after it passes

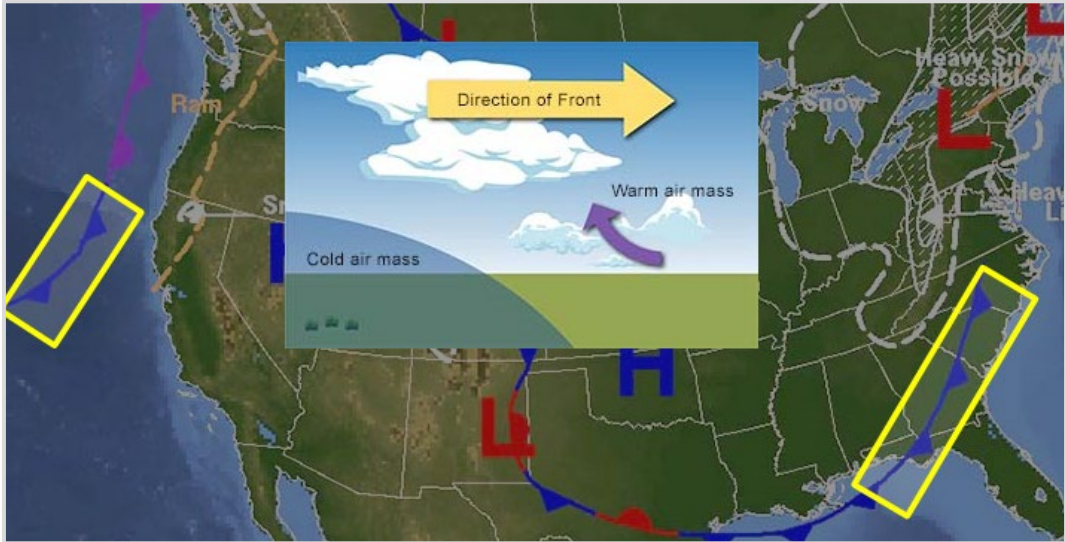
Fronts exist at the surface of the Earth and have a vertical structure in which the front slopes over the colder, denser air mass.

There are four different types of fronts:

- Cold
- Warm
- Stationary
- Occluded

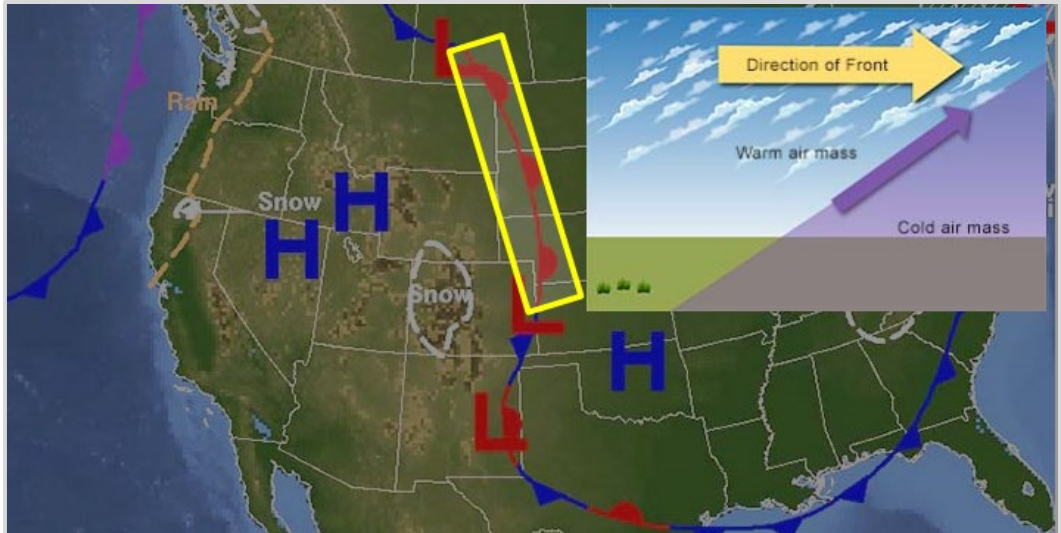


Frontal Types

TYPE OF FRONT	FEATURES/CHARACTERISTICS
Cold 	Cold fronts typically move faster than warm fronts, so in time they "catch up" to warm fronts. When this occurs, colder air replaces warmer air. 

Warm

A front that moves in such a way that warmer air replaces colder air.



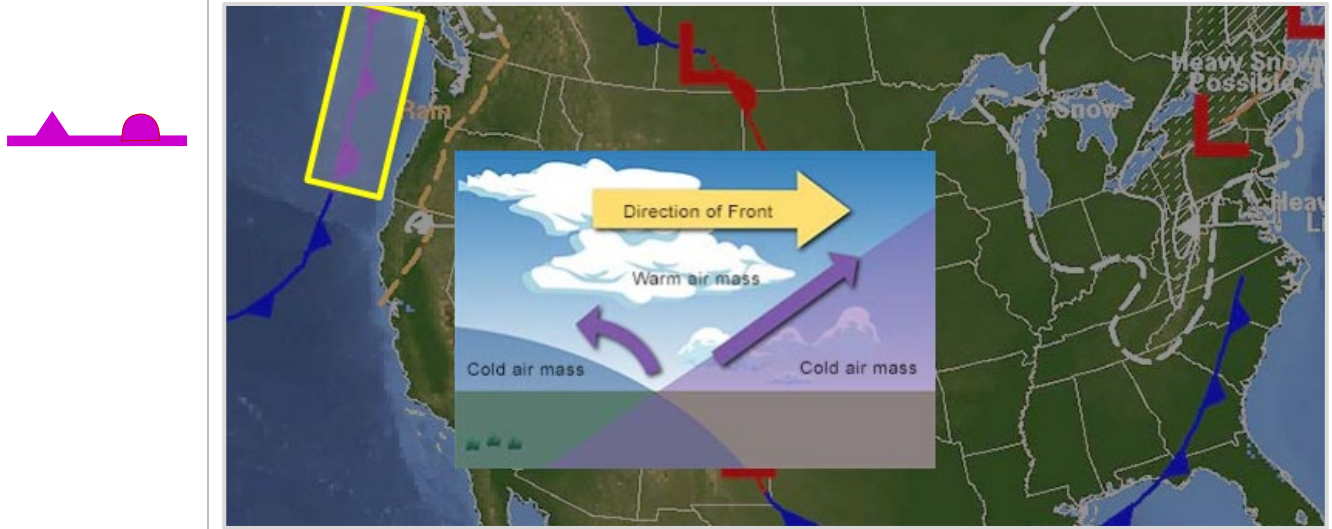
Stationary

A front which is a boundary between two different air masses. Neither air mass is strong enough to replace the other, which renders it stationary or nearly so. A wide variety of weather can be found along a stationary front, but usually, clouds and prolonged precipitation occurs here.



Occluded

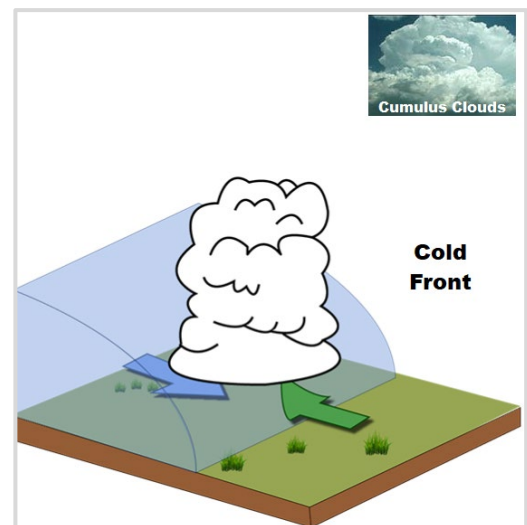
A composite of two fronts as a cold front overtakes a warm front or stationary front. The cold air undercuts the retreating cold air mass associated with the warm front, further lifting the already rising warm air. Clouds and precipitation can occur both ahead of and in the areas of frontal lift of an occluded front.



Cold Front

Cold fronts have a **steep slope** and **air is forced upward abruptly**.

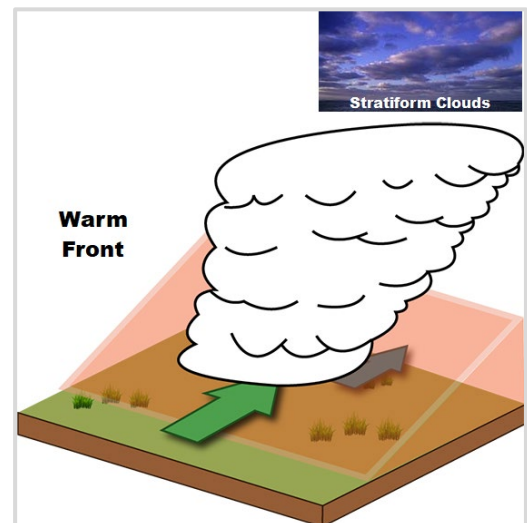
- This often leads to a narrow band of cumulus clouds, showers, and thunderstorms along or just ahead of the front if the rising air is unstable
- **Good visibility and turbulence** are associated with cold fronts



Warm Front

Warm fronts typically have a **gentle slope** so the air rising along the frontal surface is gradual.

- This favors the **development of widespread layered or stratiform cloudiness and precipitation** along and ahead of the front if the rising air is stable
- **Poor visibility, smooth air, and steady precipitation** are associated with warm fronts





Knowledge Check F

REVIEW what you have learned so far about fronts. ANSWER the questions listed below.

1. A cold air mass moving over a warm surface often produces which type of air? *(Select the correct answer.)*
 - ☐ Polluted
 - ☐ Stable
 - ☒ Unstable
2. Which front moves in such a way that warmer air replaces colder air? *(Select the correct answer.)*
 - ☐ Cold front
 - ☐ Stationary front
 - ☒ Warm front
3. What type of front has a steep slope which often leads to a narrow band of showers and thunderstorms if the rising air is unstable? *(Select the correct answer.)*
 - ☒ Cold front
 - ☐ Warm front
 - ☐ Cool front

Wind, Air Masses, and Fronts Summary

Why is weather today clear and cold over Oklahoma while it is warm and moist over Alabama? Air masses and fronts provide the answer. This lesson explained the details of air masses, mass modifications, and frontal types. You can better manage the safety and economy of air traffic when you can evaluate the expected effects of air masses and fronts.

PRECIPITATION

Purpose: This lesson will help you recognize the different ingredients and weather conditions that contribute to forms of precipitation. The lesson will also identify selected National Weather Service offices and explain their duties and responsibilities.

Objectives:

- Identify ingredients required to form precipitation
- Identify types of precipitation
- Identify duties and responsibilities of selected National Weather Service (NWS) offices

The references for this lesson are as follows:

- FAA Order JO 7110.10, Flight Services
- American Meteorology Society (AMS), Glossary of Meteorology
- FAA-H-8083-28, Aviation Weather Handbook
- NWS: Jetstream: Online School for Weather

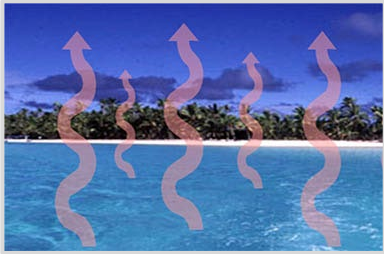
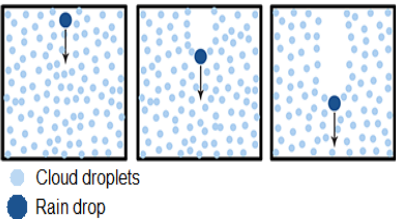
Precipitation

Precipitation is any of the forms of water particles, whether liquid or solid, that fall from the atmosphere and reach the ground.

Ingredients for Precipitation Formation

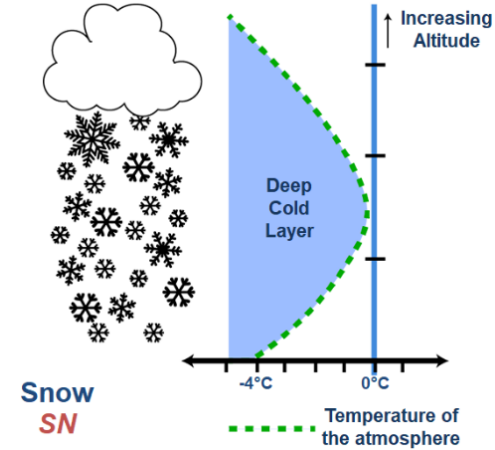
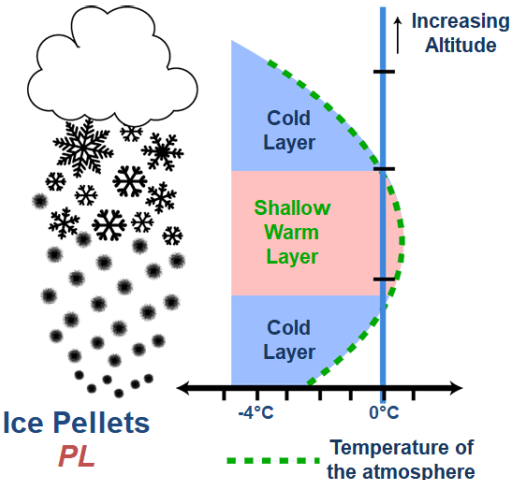
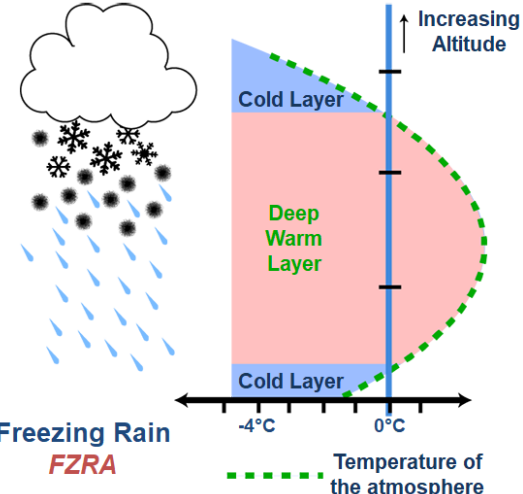


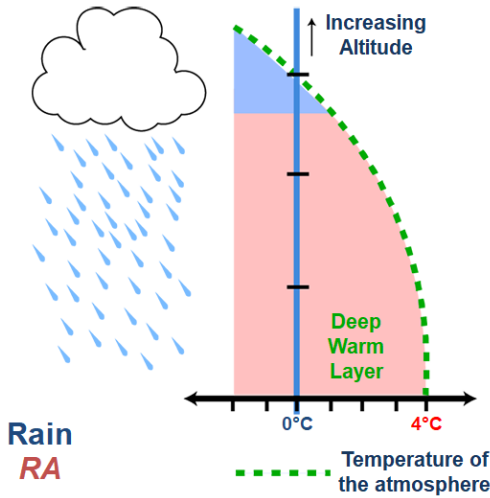
Precipitation formation requires three ingredients: water vapor, lift, and growth process.

	Water Vapor The primary sources in the United States are: ■ Atlantic and Pacific Oceans ■ Gulf of Mexico ■ Great Lakes Winds transport this water vapor inland.
	Lift Sufficient lift to condense the water vapor into clouds ■ Converging winds around surface lows ■ Fronts
	Growth Process A growth process that allows water particles to grow large and heavy enough to fall as precipitation.

Precipitation Types

The vertical distribution of temperature will often determine the type of precipitation that occurs at the surface.

TYPE OF PRECIPITATION	ILLUSTRATION
Snow (SN) – Precipitation of snow crystals, mostly branched in the form of six-pointed stars. Snow occurs when the temperature remains below-freezing throughout the entire depth of the atmosphere <i>Note: This graphic depicts an inversion; an inversion is not necessary for the development of snow.</i>	 Snow SN
Ice Pellets (PL) – Precipitation of transparent or translucent pellets of ice, which are round or irregular, rarely conical, and have a diameter of 0.2 inch (5 mm) or less. Sometimes called “sleet.” - Ice pellets occur when there is a shallow layer aloft with above-freezing temperatures, with a deep layer of below-freezing air based at the surface - As snow falls into the shallow warm layer, the snowflakes partially melt - As the precipitation re-enters air that is below freezing, it re-freezes into ice pellets	 Ice Pellets PL
Freezing Rain (FZRA) – Rain that freezes on contact with the ground or exposed objects. Freezing rain: - Occurs when there is a deep layer aloft with above-freezing temperatures, with a shallow layer of below-freezing air at the surface - Can begin as either rain or snow but becomes all rain in the warm layer - Rain falls back into below freezing air, but since the depth is shallow, it does not have time to freeze into ice pellets - Drops freeze on contact with ground or exposed objects	 Freezing Rain FZRA

TYPE OF PRECIPITATION	ILLUSTRATION
Rain (RA) – Precipitation, either in the form of drops larger than 0.02 inch (0.5 mm), or smaller drops that, in contrast to drizzle, are widely separated. Rain occurs when there is a deep layer of above-freezing air based at the surface	 The diagram shows a vertical cross-section of the atmosphere. On the left, a cloud is depicted with blue raindrops falling. To the right, a vertical axis represents altitude, with an arrow pointing upwards labeled 'Increasing Altitude'. A dashed green line represents the temperature profile of the atmosphere, which decreases as altitude increases. A solid blue line indicates the temperature at the surface is 4°C. A horizontal line at 0°C marks the freezing point. The area below 0°C is shaded pink and labeled 'Deep Warm Layer'. The area above 0°C is shaded light blue. The text 'Rain RA' is written in blue and red below the cloud.

✓ Knowledge Check G

REVIEW what you have learned so far about precipitation. ANSWER the questions listed below.

- What ingredients are needed for precipitation formation? (Select all correct answers that apply.)
 - ☒ Water vapor
 - ☐ Unstable air
 - ☒ Lift
 - ☒ Growth process
- Which precipitation type occurs when there is a deep layer aloft with above-freezing temperatures, with a shallow layer of below-freezing air at the surface? (Select the correct answer.)
 - ☐ Snow
 - ☐ Ice pellets
 - ☒ Freezing rain
 - ☐ Rain
- Which of the following conditions must be present for precipitation to occur as snow? (Select the correct answer.)
 - ☐ A deep layer of above-freezing air based at the surface.
 - ☐ A shallow layer aloft with above-freezing temperatures, with a deep layer of below-freezing air based at the surface
 - ☐ A deep layer aloft with above-freezing temperatures, with a shallow layer of below-freezing air at the surface
 - ☒ The temperature remains below freezing throughout the entire depth of the atmosphere

National Weather Service (NWS)

The National Weather Service (NWS):

- Is a federal agency under the Department of Commerce's (DOC) National Oceanic and Atmospheric Administration (NOAA)
- Provides weather, hydrologic, and climate forecasts and warnings for the protection of life and property and the enhancement of the national economy

United States NWS Offices

- Meteorological Watch Offices (MWOs)
- Weather Forecast Offices (WFOs)

National Weather Service produces the following aviation weather products:

- Airmen's Meteorological Information (AIRMETs)
- Significant Meteorological Information (SIGMETs)
- Convective SIGMETs
- Terminal Aerodrome Forecasts (TAFs)



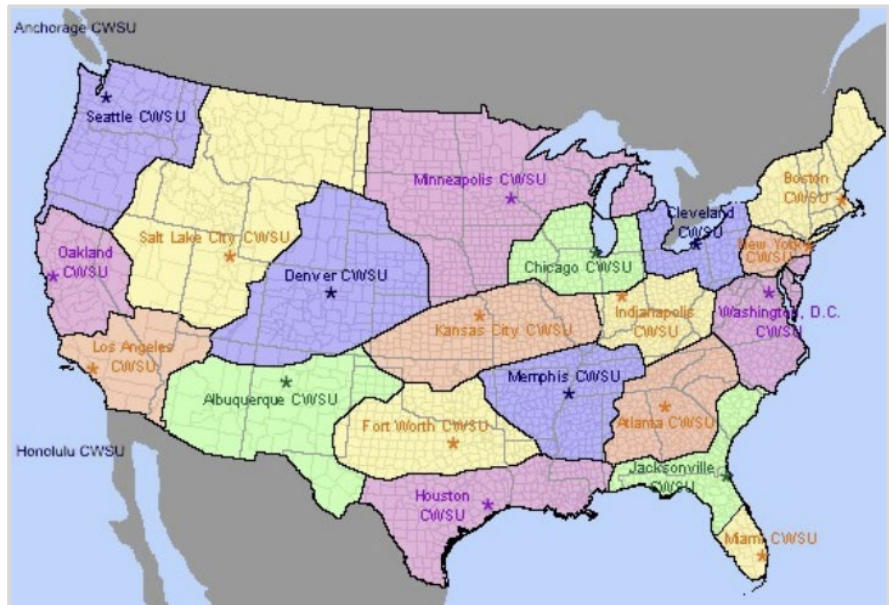
NATIONAL WEATHER SERVICE
NATIONAL OCEANIC AND ATMOSPHERIC
ADMINISTRATION

Center Weather Service Unit (CWSU)

Center Weather Service Units (CWSUs) are NWS offices located in every **Air Route Traffic Control Center (ARTCC)**, providing meteorological consultation, forecasts, and advice to ARTCCs and other FAA facilities regarding weather impact on their:

- Missions
- Equipment outages and repairs
- Staffing

Weather people



Controllers working w/ weather people

The **Center Weather Coordinator (CWC)** is staffed by controllers and is the designated interface between CWSU meteorologists and:

- ARTCC controllers
- FAA facilities within the ARTCC area of responsibility



Knowledge Check H

REVIEW what you have learned so far about National Weather Service. ANSWER the questions listed below.

1. What NWS entity provides consultation, forecast, and advice to ARTCCs regarding weather impacts? *(Select the correct answer.)*
 - ☐ Meteorological Watch Office (MWO)
 - ☐ Weather Forecast Office (WFO)
 - ☒ Center Weather Service Unit (CWSU)
2. The CWSU provides information to ARTCCs and other FAA facilities regarding weather impact for which of the following areas? *(Select all correct answers that apply.)*
 - ☒ Equipment outages and repairs
 - ☒ Missions
 - ☐ Runway status
 - ☒ Staffing
3. Which organization is responsible for coordination between the CWSU meteorologists and ARTCC controllers and facilities? *(Select the correct answer.)*
 - ☐ MWO
 - ☐ CWSU
 - ☐ ARTCC
 - ☒ CWC

Precipitation Summary

Different types of precipitation can affect how an aircraft functions, which can be detrimental to the safety of pilots, passengers, and crew on an aircraft. As an air traffic controller, you must be advised of all current weather conditions that affect or could potentially affect your location.

You are responsible for obtaining updated and accurate forecasts or advisories from reliable resources. Being advised of weather conditions is a vital resource to controllers, as they strive to maintain the safety of those who rely on their guidance and direction.

SUMMARY

The purpose of this module was to explain the basic principles of weather, different elements of the atmosphere, and the National Weather Service's (NWS) role in providing aviation weather services to the FAA.

In accordance with FAA Order JO 7110.10, Flight Services; NWS: Jetstream: Online School for Weather; FMH-1, Surface Weather Observations and Reports; FAA-H-8083-28, Aviation Weather Handbook; and the American Meteorology Society (AMS), Glossary of Meteorology, you should now be able to:

- Identify characteristics and properties of the Earth's atmosphere
- Describe characteristics of atmospheric moisture
- Define forms of clouds
- Identify air masses and their characteristics
- Identify air mass modifications and their characteristics
- Identify frontal types and their characteristics
- Identify ingredients required to form precipitation
- Identify types of precipitation
- Identify duties and responsibilities of selected National Weather Service (NWS) Offices

KNOWLEDGE CHECK ANSWER KEY

Knowledge Check A

1. Troposphere; 2. Stratosphere; 3. 29.92 inches of mercury; 4. Jet streams; 5. Polar and Subtropical

Knowledge Check B

1. Sublimation; 2. Relative humidity; 3. Saturation

Knowledge Check C

1. Rising; 2. Cumuliform; 3. Stratiform; 4. Cumulonimbus, Nimbostratus

Knowledge Check D

1. Low; 2. High

Knowledge Check E

1. Warm, moist; 2. Turbulence

Knowledge Check F

1. Unstable; 2. Warm front; 3. Cold front

Knowledge Check G

1. Water vapor, Lift, Growth process; 2. Freezing rain; 3. The temperature remains below freezing throughout the entire depth of the atmosphere

Knowledge Check H

1. Central Weather Service Unit (CWSU); 2. Equipment outages and repairs, Missions, Staffing; 3. CWC